

About This Document

ML, model-evaluation and deep-learning MCQs are reported across the assessment and interview.

Focus areas (research-backed): Bias-variance tradeoff and R-squared vs adjusted R-squared are explicitly reported in Tredence interviews.

Core Machine Learning Concepts

14 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Overfitting occurs when a model:

(A) Performs poorly on both train and test

(B) Performs well on train but poorly on unseen data ✓

(C) Has too few parameters

(D) Cannot be trained

Answer: (B) Performs well on train but poorly on unseen data

Authenticity: Frequently Repeated

Q2. Which technique reduces overfitting?

(A) Increasing complexity

(B) Regularization ✓

(C) Removing validation

(D) Fewer epochs only

Answer: (B) Regularization

Authenticity: Frequently Repeated

Q3. Supervised learning requires:

(A) Unlabeled data

(B) Labeled data ✓

(C) No data

(D) Only test data

Answer: (B) Labeled data

Authenticity: Frequently Repeated

Q4. Which is an unsupervised algorithm?

(A) Logistic Regression

(B) K-means clustering ✓

(C) Decision Tree classifier

(D) Linear Regression

Answer: (B) K-means clustering

Authenticity: Frequently Repeated

Q5. In the bias-variance tradeoff, high bias leads to:

(A) Overfitting

(B) Underfitting ✓

(C) Perfect fit

(D) Data leakage

Answer: (B) Underfitting

Authenticity: Community Reported

Q6. Random Forest is an ensemble of:

(A) K-means runs

(B) Decision trees ✓

(C) Neural nets

(D) SVMs

Answer: (B) Decision trees

Authenticity: Community Reported

Q7. PCA is primarily used for:

(A) Classification

(B) Dimensionality reduction ✓

(C) Clustering labels

(D) Labeling

Answer: (B) Dimensionality reduction

Authenticity: Community Reported

Q8. Which is NOT a supervised algorithm?

(A) Linear Regression

(B) Logistic Regression

(C) K-means ✓

(D) Random Forest

Answer: (C) K-means

Authenticity: Frequently Repeated

Q9. Boosting is an ensemble method that:

(A) Trains models independently in parallel

(B) Builds models sequentially, each correcting prior errors ✓

(C) Reduces features

(D) Clusters data

Answer: (B) Builds models sequentially, each correcting prior errors

Authenticity: Community Reported

Q10. Feature scaling matters most for:

(A) Tree-based models

(B) Distance- and gradient-based algorithms ✓

(C) Rule systems

(D) Tokenizers

Answer: (B) Distance- and gradient-based algorithms

Authenticity: Community Reported

Q11. The Naive Bayes classifier assumes:

(A) Features are dependent

(B) Conditional independence of features ✓

(C) Linear boundaries

(D) No priors

Answer: (B) Conditional independence of features

Authenticity: Community Reported

Q12. KNN is best described as a:

(A) Lazy learner (no explicit training) ✓

(B) Deep model

(C) Linear model

(D) Generative model

Answer: (A) Lazy learner (no explicit training)

Authenticity: Community Reported

Q13. A common splitting criterion in decision trees is:

(A) Mean squared padding

(B) Gini impurity or entropy ✓

(C) Softmax

(D) L2 norm

Answer: (B) Gini impurity or entropy

Authenticity: Community Reported

Q14. SVM aims to:

(A) Minimize the number of features

(B) Maximize the margin between classes ✓

(C) Cluster points

(D) Reduce dimensions

Answer: (B) Maximize the margin between classes

Authenticity: Community Reported

Model Evaluation

10 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Best metric for an imbalanced classification problem:

(A) Accuracy

(B) F1-score ✓

(C) Total count

(D) Learning rate

Answer: (B) F1-score

Authenticity: Frequently Repeated

Q2. Precision is:

(A) $TP/(TP+FN)$

(B) $TP/(TP+FP)$ ✓

(C) $TN/(TN+FP)$

(D) $TP/Total$

Answer: (B) $TP/(TP+FP)$

Authenticity: Frequently Repeated

Q3. Recall (sensitivity) is:

(A) $TP/(TP+FP)$

(B) $TP/(TP+FN)$ ✓

(C) $TN/(TN+FN)$

(D) $FP/(FP+TN)$

Answer: (B) $TP/(TP+FN)$

Authenticity: Frequently Repeated

Q4. A confusion matrix shows:

(A) Feature correlations

(B) Counts of TP/TN/FP/FN ✓

(C) Model weights

(D) Learning curves

Answer: (B) Counts of TP/TN/FP/FN

Authenticity: Community Reported

Q5. Cross-validation is used to:

(A) Increase data size

(B) Estimate generalization to unseen data ✓

(C) Reduce classes

(D) Encode features

Answer: (B) Estimate generalization to unseen data

Authenticity: Community Reported

Q6. Purpose of a train-test split:

(A) Speed up training

(B) Evaluate on unseen data ✓

(C) Balance classes

(D) Remove outliers

Answer: (B) Evaluate on unseen data

Authenticity: Frequently Repeated

Q7. Typical loss for regression:

(A) Binary cross-entropy

(B) Mean Squared Error ✓

(C) Hinge loss

(D) Categorical cross-entropy

Answer: (B) Mean Squared Error

Authenticity: Community Reported

Q8. Typical loss for binary classification:

(A) MSE

(B) Binary cross-entropy ✓

(C) MAE

(D) R-squared

Answer: (B) Binary cross-entropy

Authenticity: Community Reported

Q9. Adjusted R-squared differs from R-squared because it:

(A) Always increases with more features

(B) Penalizes adding non-useful predictors ✓

(C) Ignores residuals

(D) Equals R-squared always

Answer: (B) Penalizes adding non-useful predictors

Authenticity: Community Reported

Q10. The bias-variance tradeoff describes the balance between:

(A) Speed and memory

(B) Underfitting and overfitting ✓

(C) Precision and recall

(D) Train and test size

Answer: (B) Underfitting and overfitting

Authenticity: Community Reported

Deep Learning & Preprocessing

10 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Common hidden-layer activation in deep networks:

(A) Sigmoid

(B) ReLU ✓

(C) Linear

(D) Step

Answer: (B) ReLU

Authenticity: Community Reported

Q2. The sigmoid function outputs values in:

(A) -1 to 1

(B) 0 to 1 ✓

(C) 0 to inf

(D) -inf to inf

Answer: (B) 0 to 1

Authenticity: Community Reported

Q3. Output activation for multi-class classification:

(A) ReLU

(B) Softmax ✓

(C) Tanh

(D) Linear

Answer: (B) Softmax

Authenticity: Community Reported

Q4. Gradient descent is used to:

(A) Increase the loss

(B) Minimize the cost function ✓

(C) Split data

(D) Encode features

Answer: (B) Minimize the cost function

Authenticity: Frequently Repeated

Q5. If the learning rate is too high, training may:

(A) Converge faster with no downside

(B) Fail to converge and overshoot ✓

(C) Always reach global minimum

(D) Stop updating

Answer: (B) Fail to converge and overshoot

Authenticity: Community Reported

Q6. One-hot encoding handles:

(A) Missing numerics

(B) Categorical variables ✓

(C) Outliers

(D) Imbalance

Answer: (B) Categorical variables

Authenticity: Frequently Repeated

Q7. A common way to handle missing values is:

(A) Encoding

(B) Imputation ✓

(C) Normalization

(D) Tokenization

Answer: (B) Imputation

Authenticity: Community Reported

Q8. L1 (Lasso) regularization tends to produce:

(A) Dense weights

(B) Sparse models (some weights zero) ✓

(C) Larger weights

(D) No change

Answer: (B) Sparse models (some weights zero)

Authenticity: Community Reported

Q9. Dropout in a neural network is used to:

(A) Speed up I/O

(B) Reduce overfitting ✓

(C) Increase parameters

(D) Normalize inputs

Answer: (B) Reduce overfitting

Authenticity: Community Reported

Q10. One epoch means:

(A) One weight update

(B) One full pass over the training data ✓

(C) One batch

(D) One layer

Answer: (B) One full pass over the training data

Authenticity: Community Reported

Tredence

Machine Learning & Deep Learning

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.